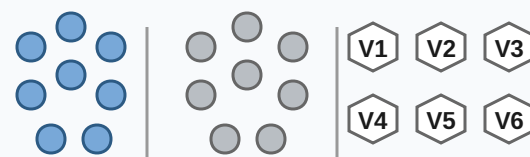


Pilot screen

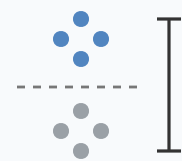
- 25-50 cells per perturbation
- balanced WT / control
- initial variant panel



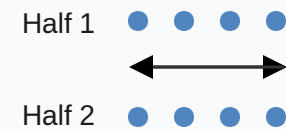
Estimate measurement properties

- variant-to-WT effect size
- within-variant split-half noise
- pilot effect-size-to-noise estimate

Variant vs WT

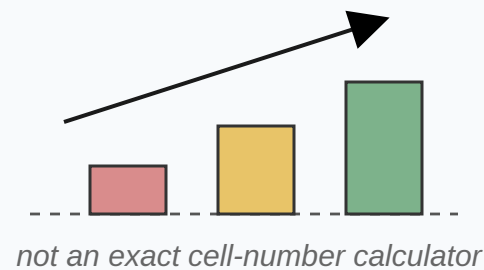


Split-half (same variant)



Predict rankability

- predicted rankability
- expected measurement regime
- relative triage priority



Predicted to
clear the
measurement
floor?

Yes

No

Proceed to full-depth benchmarking

- size experiment to target depth
- evaluate direction, PDS and DE fidelity
- compare models on measurable perturbations



Redesign or classify as measurement-limited

- increase depth where plausible
- enrich stronger phenotypes or conditions
- change assay, stimulation or cell state
- improve variant assignment or controls



do not interpret model ranking below the floor

Report benchmark by measurement regime



Measurable / rankable perturbations

- model performance
- benchmark ranking
- uncertainty



Below-floor / measurement-limited perturbations

- report separately
- do not label as model failure
- document redesign needs



Benchmark models only where the ground truth is sufficiently resolved.

Use rankability as a relative triage tool, not a universal depth calculator.

re-pilot after redesign